



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)
Mid Term-I (Session: 2025-26)

Roll No.:

Max. Marks: 15
Time: 75 Minutes
Semester: I

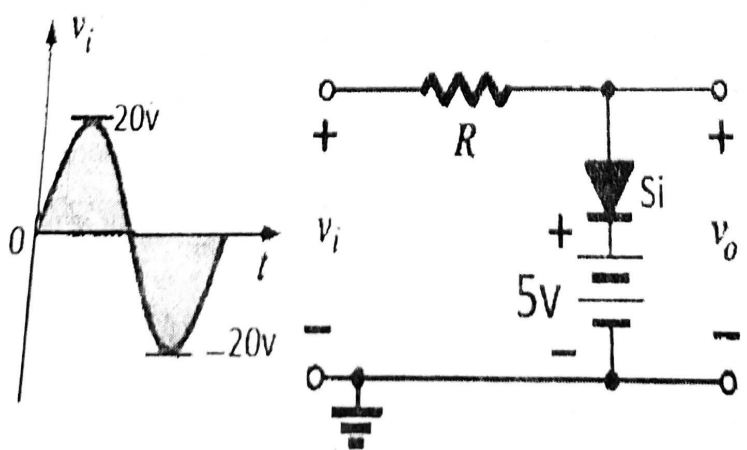
Course Name: B.Tech
Subject Name: Basics Electronics Engineering
Subject Code: 25EC101

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
(1 x 5)				
1. Attempt All Questions				
(a)	Define barrier potential of a P-N junction diode.	CO1	K2	2.4.4
(b)	Define Peak inverse voltage (PIV). Write the PIV rating of Centre tap transformer type full wave rectifier.	CO1	K2	2.3.1
(c)	Define ripple factor and write the values for the half wave and full wave rectifiers.	CO1	K2	2.3.1
(d)	BJT is bipolar or unipolar? Justify your answer.	CO2	K2	15.1.1
(e)	Why the size of collector is larger than emitter?	CO2	K2	1.3.1

SECTION-B (Minimum Knowledge Level: K2)				
(Same K Levels) (2 x 2)				
2. Attempt Any two Question				
(a)	Draw the V-I characteristics of p-n junction diode. Also define breakdown voltage.	CO1	K2	16.1.1
(b)	Explain the working of Centre tap transformer type Full wave rectifier.	CO1	K2	5.1.2
(c)	Explain the application of Zener diode as a voltage regulator.	CO1	K2	16.1.1

SECTION-C (Minimum Knowledge Level: K3)				
(Same K Levels) (3 x 1)				
3. Attempt Any one Question				
(a)	Determine and draw the output of the network given below. 	CO1	K3	2.1.3

Determine and draw the output of the network given below.			
(b)		CO1	K3
			2.1.3

4. Attempt Any one Question

(Same K Levels) (3 x 1)

(a)	Explain the working of N-P-N transistor. If the value of α is 0.988 then determine the value of I_C ($I_E = 10.2$ milliampere).	CO2	K3	15.1.1
(b)	Draw and explain the input and output characteristics of Common base configuration. If $I_C = 10$ milliampere & $I_E = 10.2$ milliampere then determine the value of α .	CO2	K3	15.1.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

SECTION-A

Minimum Knowledge Level: K2

MT 02

Q. No.

1. Attempt All Questions

Q. No.	Question	CO	KL	MT
(a)	Find 2's complement of the given number 101101.	CO3	K3	1.4.1
(b)	State De Morgan's Theorem.	CO3	K2	1.5.1
(c)	Consider $X = 1010$; $Y = 1000$: Perform $X+Y$.	CO3	K3	1.5.1
(d)	What is Pinch off Condition in FET?	CO2	K2	1.4.1
(e)	How JFET can be used as Voltage Variable resistor?	CO2	K2	1.4.1

SECTION-B (Minimum Knowledge Level: K2)

2. Attempt Any two Question

(Same K Levels) (2 x 2)

(a)	Realize the NOT, AND and OR gates using NAND gate.	CO3	K3	1.5.1
(b)	Perform the following: i) $(73.12)_{10} = ()_{16}$ ii) $(750.6032)_8 = ()_{16}$	CO3	K3	1.5.1
(c)	Find the minterms and maxterms for the given expression: $Y = (\bar{A} + B)(\bar{B} + C)$	CO3	K3	1.4.1

SECTION-C (Minimum Knowledge Level: K3)

3. Attempt Any one Question

(Same K Levels) (3 x 1)

(a)	Simplify the expression using K-Map and implement using basic gates. $Y = \prod M(0,1,2,3,7,8,10) + d(5,6,11,15)$	CO3	K3	1.4.1
(b)	Simplify the expression using K-Map and implement using basic gates. $F(A, B, C, D) = \sum m(0,1,4,6,8,9,11,12,14) + d(3,7)$	CO3	K3	1.4.1

4. Attempt Any one Question

(Same K Levels) (3 x 1)

(a)	Explain the working of Depletion type MOSFET.	CO2	K2	1.4.1
(b)	Explain the working of CMOS as an Inverter.	CO2	K2	1.4.1



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Pre - End Term Exam (Session: 2025-26)

Admission No.: [REDACTED]

Course Name: B. Tech

Subject Name: Basics of Electronics Engineering

Subject Code: 25EC101

Max. Marks: 20

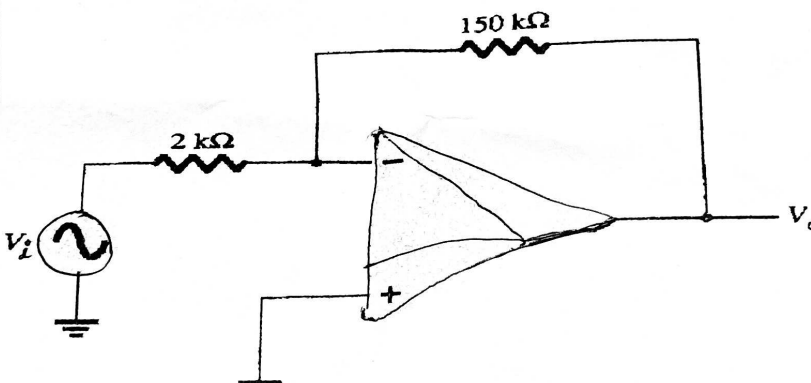
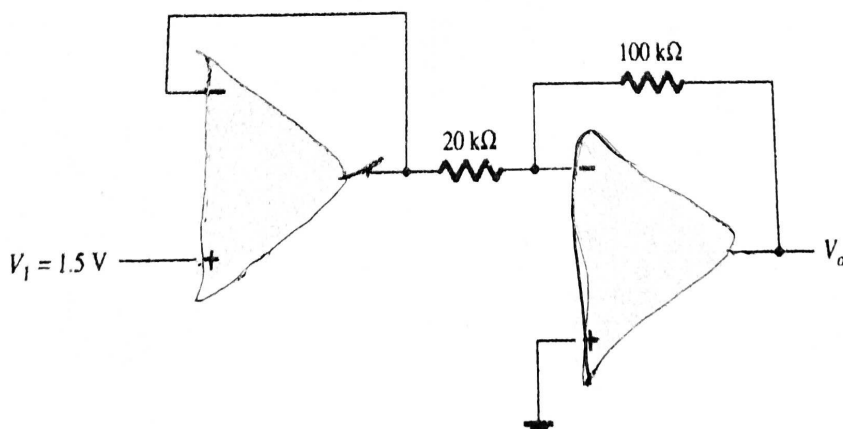
Time: 90 Minutes

Semester: 1st

Note:

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3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Knowledge Level: K2 & K3)

Q. No.	Question	CO	KL	PI
Attempt any one part from each question (3 Marks x 4 Questions = 12 Marks)				
1.(a)	Write the ideal characteristics of an op-amp (any six) and design an inverting amplifier having voltage gain of 50.	CO4	K3	2.1.3
1.(b)	Derive the expression of output voltage of a non-inverting Op-amp.	CO4	K3	2.1.3
2.(a)	Determine output voltage of the following network ($V_i = 100\text{mV}$). 	CO4	K3	1.1.2
2.(b)	Determine output voltage of the following network. 	CO4	K3	1.1.2
3.(a)	Write comparisons amongst 1G, 2G, 3G, 4G and 5G wireless communication technologies.	CO5	K2	5.1.1

3.(b)	Explain the various elements of a communication system.	CO5	K2	5.1.1
4.(a)	Explain Wireless Sensor Network (WSN) using suitable architecture.	CO5	K2	2.1.1
4.(b)	Differentiate between Sensor & Transducer.	CO5	K2	2.1.1
SECTION-B (Knowledge Level: K3)				
Attempt any one part from each question (4 Marks x 2 Questions = 8 Marks)				
5.(a)	Draw & explain the working of IC-555 Timer in Astable mode using suitable circuit diagram.	CO4	K2	3.1.1
5.(b)	Draw & explain the working of IC-555 Timer in Mono-stable mode using suitable circuit diagram.	CO4	K2	3.1.1
6.(a)	Define modulation & its need in detail.	CO5	K2	5.1.4
6.(b)	Write a short note on IoT architecture.	CO5	K2	5.1.4

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

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